



Nottingham Trent
University

Department of Computer Science

COMP40731 – Artificial Cognitive Systems

Lecture 7: Cortical and Noisy Neurons

Pedro Machado pedro.machado@ntu.ac.uk



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University

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Lecture Outline

In this lecture will cover:

- Cortical Neurons Properties
- Suitable Neurons for Cortical Responses
- Spike Train Variability
- Are Neurons Noisy?
- Noise Sources
- Statistics of Spike Trains

This lecture contributes to the following learning outcomes:

MLO2 - Demonstrate awareness of the different main paradigms corresponding to each of the several disciplines involved (computer science, digital electronics, AI), to be able to contextualise cognitive systems research and its current outlook.

MLO4. Understand and explain the applications areas of ACS, and the consequent social and ethical issues.

MLO6. Demonstrate in-depth skills in system architecture design and programming required to implement ACS.

MLO08. Demonstrate the capability of testing applications targeting real applications.

MLO10. Demonstrate the use of Distributed Database systems to handle data scalability and considering costs, benefits, risks and relevant business goals.

- Escape Noise

Cortical Spiking Neurons

- Understanding the dynamics of cortical spiking neurons is crucial for unraveling the mysteries of information processing in the brain.
- These neurons play a fundamental role in various cognitive processes, including perception, learning, and memory.
- Choosing the right model is essential for accurately capturing the complex behavior of cortical spiking neurons.
- Let's explore different computational models commonly used in neuroscience and evaluate their suitability for simulating cortical spiking neurons.
- Various types of models are employed to simulate cortical spiking neurons, catering to different research needs and computational capabilities.
- Simple integrate-and-fire models offer computational efficiency but may lack biophysical realism, while complex models like Hodgkin–Huxley provide detailed physiological insights but demand significant computational resources.
- Each model type has its own set of advantages and limitations, highlighting the importance of selecting the appropriate model based on research objectives and available computational resources

Neuro-Computational Features

A. Tonic Spiking

- Neurons exhibit tonic spiking behavior when stimulated with direct current (dc), resulting in a continuous train of spikes, as depicted in Fig. 1(a).
- This firing pattern is characteristic of various types of cortical neurons, including regular spiking (RS) excitatory neurons, low-threshold spiking (LTS) neurons, and fast spiking (FS) inhibitory neurons.
- Tonic spiking indicates the presence of persistent input, as the neuron maintains continuous firing as long as the input is applied.

B. Phasic Spiking

- In contrast to tonic spiking, phasic spiking involves the neuron firing only a single spike at the onset of the input, as shown in Fig. 1(b).
- Following the initial spike, the neuron remains quiescent, making phasic spiking useful for detecting the beginning of stimulation.

C. Tonic Bursting

- Some neurons, such as chattering neurons in the cat neocortex, exhibit tonic bursting behavior characterized by periodic bursts of spikes in response to stimulation, as illustrated in Fig. 1(c).
- The interburst frequency of these neurons can be as high as 50 Hz, contributing to gamma-frequency oscillations in the brain.

D. Phasic Bursting

- Similar to phasic spiking, phasic bursting neurons respond to stimulation with a burst of spikes followed by quiescence, as depicted in Fig. 1(d).
- These bursts are believed to convey the saliency of the input and facilitate selective communication between neurons through encoded interspike frequencies.

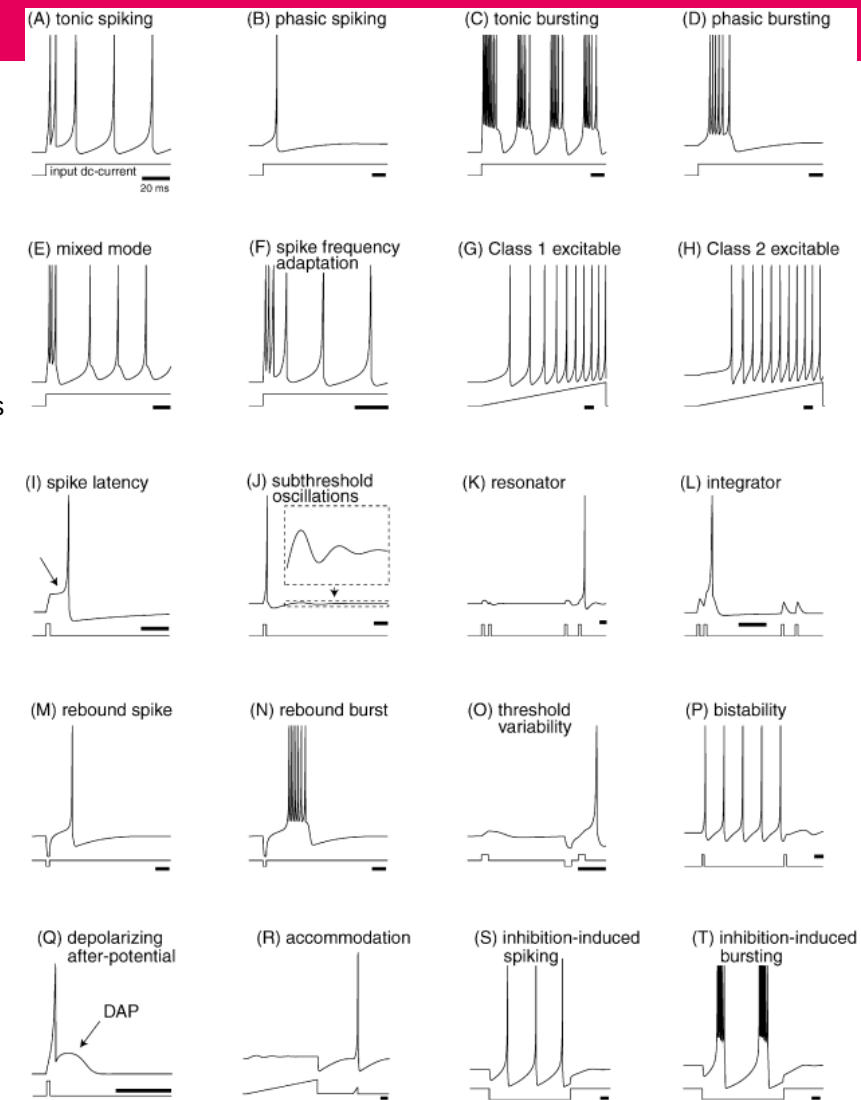


Fig. 1. Summary of the neuro-computational properties of biological spiking neurons. Shown are simulations of the same model (1) and (2), with different choices of parameters. Each horizontal bar denotes a 20-ms time interval. (Izhikevich, 2004)

Neuro-Computational Features

E. Mixed Model (Bursting Then Spiking)

- Intrinsically bursting (IB) excitatory neurons in the mammalian neocortex display a mixed spiking pattern, as shown in Fig. 1(e).
- Initially, they fire a phasic burst in response to stimulation before transitioning to tonic spiking mode.
- The computational significance of this mixed activity, beyond detecting stimulus onset, is not fully understood.

F. Spike Frequency Adaptation

- Regular spiking (RS) cells, common in the mammalian neocortex, exhibit tonic spiking with decreasing frequency over time, as depicted in Fig. 1(f).
- This adaptation in spike frequency, observed also in low-threshold spiking (LTS) inhibitory neurons, may encode the duration of stimulus onset.

G. Class 1 Excitability

- Neocortical RS excitatory neurons demonstrate Class 1 excitability, where their tonic spiking frequency varies with input strength, spanning a wide range from 2 Hz to 200 Hz or more (Fig. 1(g)).
- This ability allows them to encode input strength into their firing rate.

H. Class 2 Excitability

- Some neurons, known as Class 2 excitable, exhibit either quiescence or fire at a relatively high, constant frequency, such as 40 Hz (Fig. 1(h)).
- Unlike Class 1 excitable, their firing rate poorly predicts input strength.

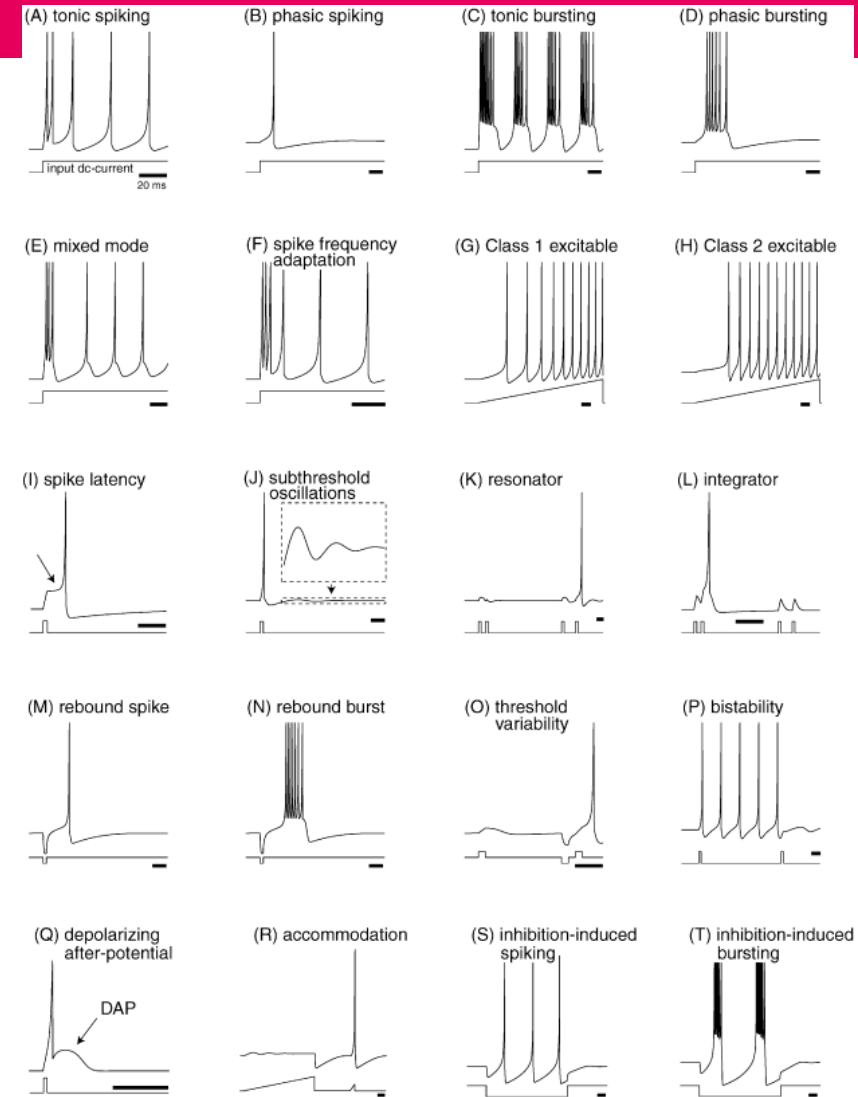


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Neuro-Computational Features

I. Spike Latency

- Cortical neurons typically fire spikes with a delay, or spike latency, dependent on input strength (Fig. 1(i)).
- This latency mechanism provides a spike-timing mechanism to encode input strength.

J. Subthreshold Oscillations

- Neurons across brain structures can generate oscillatory potentials, serving as bandpass filters (Fig. 1(j)).
- The frequency of these oscillations is crucial for information processing in the brain.

K. Frequency Preference and Resonance

- Neurons with oscillatory potentials can selectively respond to inputs with frequencies similar to their subthreshold oscillations due to resonance (Fig. 1(k)).
- This phenomenon allows for frequency-modulated interactions and signal multiplexing, making these neurons effective resonators.

L. Integration and Coincidence Detection

- Neurons without oscillatory potentials function as integrators, exhibiting a preference for high-frequency inputs (Fig. 1(l)).
- Higher input frequencies increase the likelihood of firing, facilitating the detection of coincident or nearly coincident spikes.

M. Rebound Spike

- Following the release from inhibitory input, neurons may fire a post-inhibitory (rebound) spike, as depicted in Fig. 1(m).
- This rebound spike phenomenon, akin to anodal break excitation in excitable membranes, blurs the distinction between excitation and inhibition in many spiking neurons.

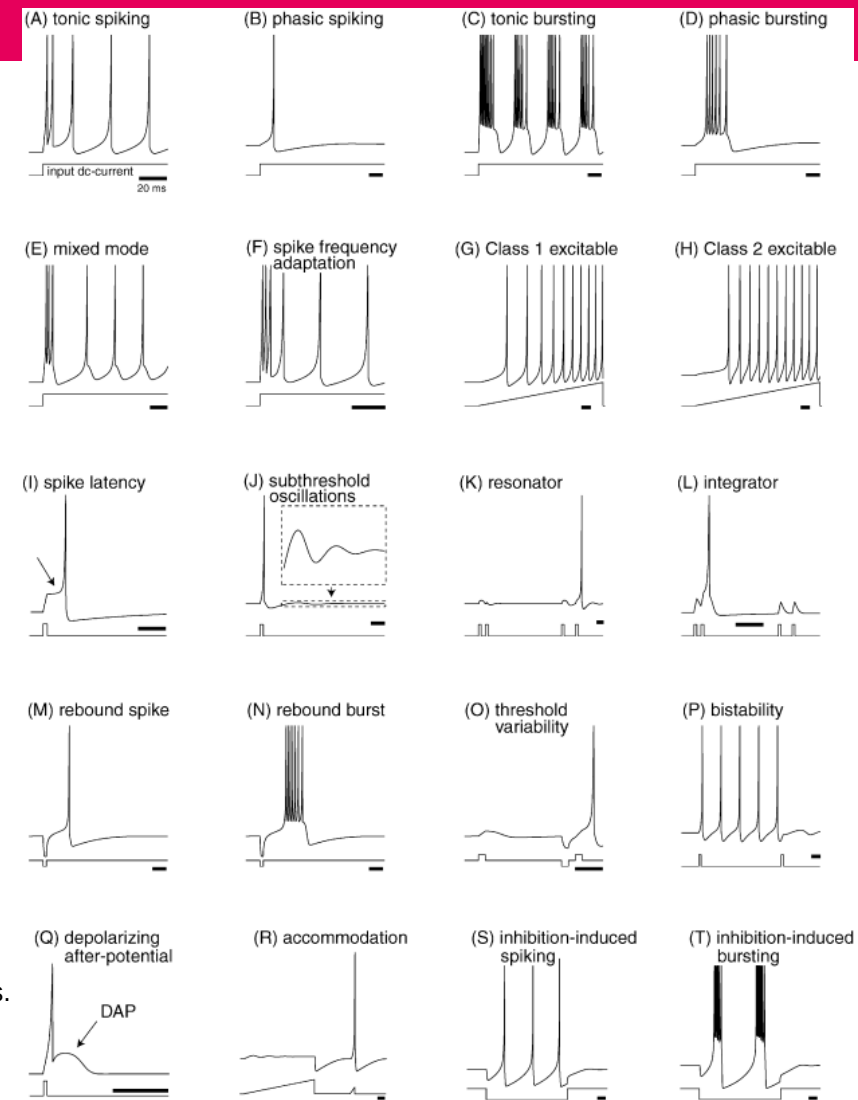


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Neuro-Computational Features

N. Rebound Burst

- Certain neurons, like thalamo-cortical cells, can produce post-inhibitory bursts (Fig. 1(n)), contributing to sleep oscillations in the thalamo-cortical system.

O. Threshold Variability

- Biological neurons exhibit variable thresholds influenced by prior activity (Fig. 1(o)).
- Preceding inhibitory input can lower the threshold, making a previously subthreshold depolarization now trigger firing.

P. Bistability of Resting and Spiking States

- Some neurons show bistability, transitioning between resting and tonic spiking (or bursting) states with excitatory or inhibitory pulses (Fig. 1(p)).
- Timing of input arrival, relative to the oscillatory phase, is crucial for mode switching, highlighting the significance of spike-timing in information processing.

Q. Depolarising After-Potentials

- Neurons may display prolonged depolarized after-potentials following a spike, affecting their excitability (Fig. 1(q)).
- Factors contributing to depolarizing after-potentials include dendritic influence, high-threshold inward currents, or interplay between subthreshold voltage-gated currents.

R. Accommodation

- Neurons exhibit accommodation, where they may not fire in response to slowly increasing input but are sensitive to brief coincident inputs (Fig. 1(r)).

During slow input ramps, inward currents may inactivate and outward currents activate, reducing excitability and preventing spike generation.

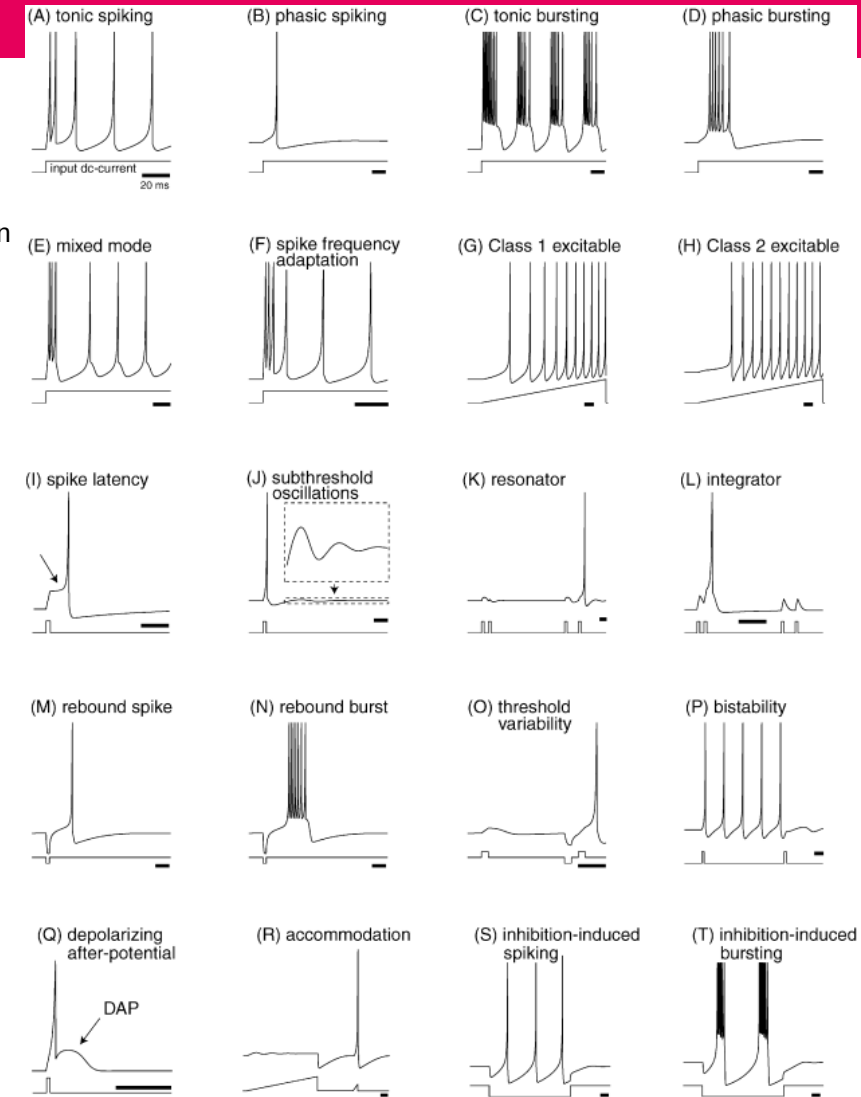


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Neuro-Computational Features

S. Inhibition-Induced Spiking

- Thalamo-cortical neurons can remain quiet without input but fire when hyperpolarised by inhibition or injected current (Fig. 1(s)).
- This response occurs due to current-induced activation of the h-current and deinactivation of calcium T-current, leading to tonic spiking.

T. Inhibition-Induced Bursting

- Thalamo-cortical neurons can also burst with spikes instead of single spikes when hyperpolarised for an extended duration (Fig. 1(t)).
- This bursting pattern is associated with spindle wave oscillations in the thalamo-cortical system, contributing to sleep rhythms.
- It's important to note that no single model can exhibit all 20 neurocomputational properties simultaneously because some are mutually exclusive.
- However, certain models can be adjusted to demonstrate each property individually.
- For example, the neuronal responses in Fig. 1 were produced using a simple spiking model with four easily adjustable parameters

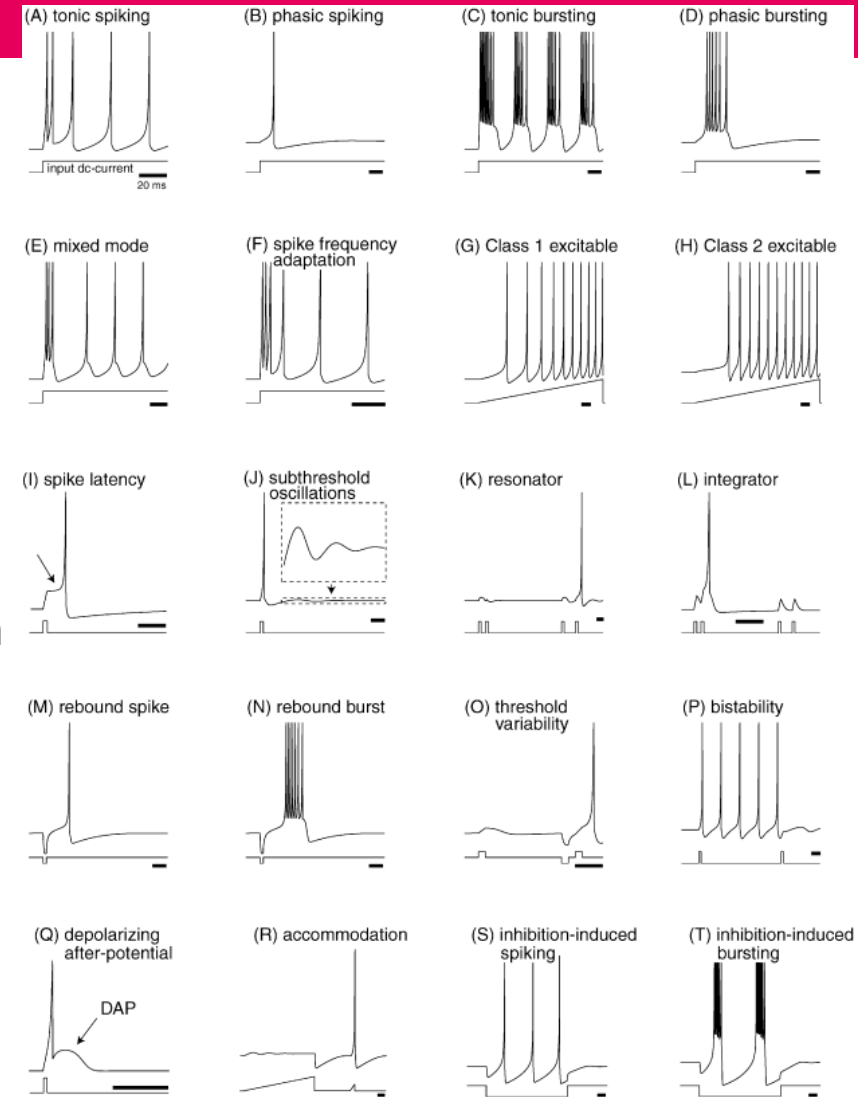


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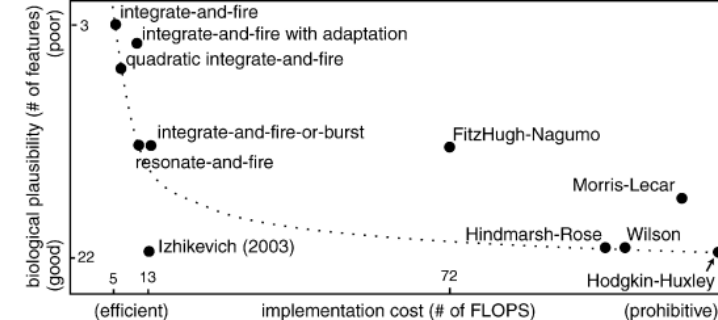
Spiking neuron models

A. Leaky Integrate-and-Fire (I&F)

- The leaky integrate-and-fire (I&F) neuron is widely used in computational neuroscience.
- Its dynamics are described by the equation

$$v' = I + a - bv, \text{ if } v \geq v_{\text{thresh}}, \text{ then } v \leftarrow c$$

- where v is the membrane potential, I is the input current, and a , b , c , and v_{thresh} are parameters.
- When v reaches the threshold v_{thresh} , the neuron fires a spike, resetting v to c .
- The I&F neuron is Class 1 excitable, firing tonic spikes with constant frequency and acting as an integrator.
- It's simple to implement with an integration time step τ of 1 ms, requiring minimal computational resources.
- However, it lacks features like phasic spiking, bursting, rebound responses, threshold variability, bistability of attractors, or chaotic dynamics.
- The fixed threshold means spikes don't have latencies, making it less suitable for modeling certain neuronal behaviors.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	spike latency	subthreshold oscillations	resonator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
integrate-and-fire	-	+	-	-	-	-	+	-	-	-	-	+	-	-	-	-	-	-	-	-	5
integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	10
integrate-and-fire-or-burst	-	+	+	+	-	+	+	-	-	-	-	+	+	+	-	+	+	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	-	+	+	+	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	72
Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

Spiking neuron models

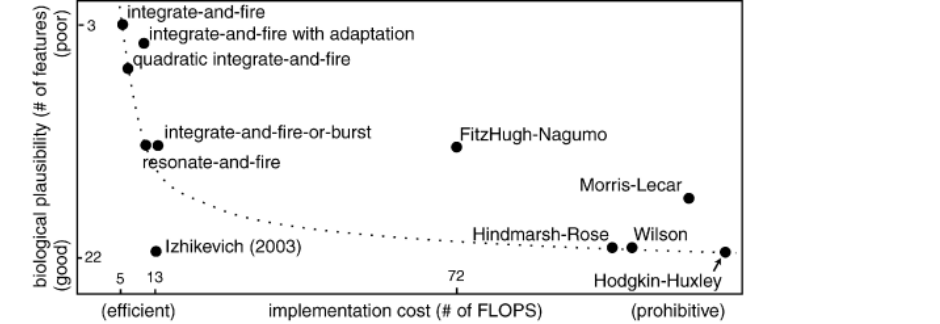
B. I&F with Adaptation

- The basic I&F model lacks bursting and other cortical neuron properties due to its one-dimensional nature.

$$v' = I + a - bv + g(d - v)$$

$$g' = \frac{(e\delta(t) - g)}{\tau}$$

- Adding a second linear equation to describe activation dynamics of a high-threshold K current can introduce spike-frequency adaptation.
- The additional equation modifies the membrane potential dynamics, increasing K activation after each firing through a Dirac delta function.
- This adaptation mechanism leads to a slowing down of the frequency of tonic spiking over time.
- Despite this enhancement, the model still falls short in replicating many crucial properties of cortical spiking neurons.
- Simulations of this adapted model require 10 floating-point operations per 1 ms time step, maintaining computational efficiency but sacrificing biological fidelity.



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Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
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Spiking neuron models

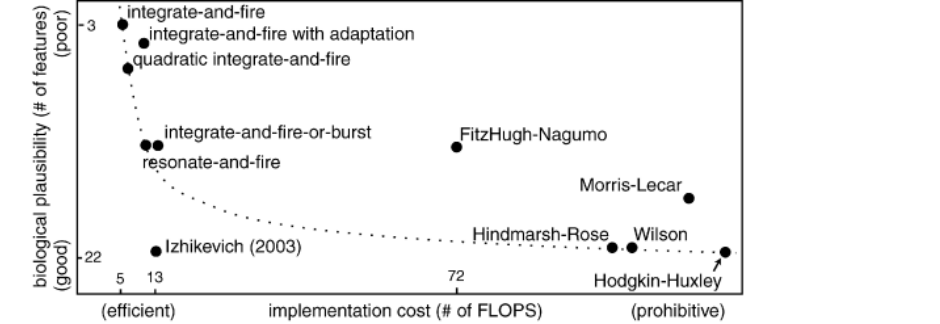
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Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	-	72
Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

Spiking neuron models

C. Integrate-and-Fire-or-Burst

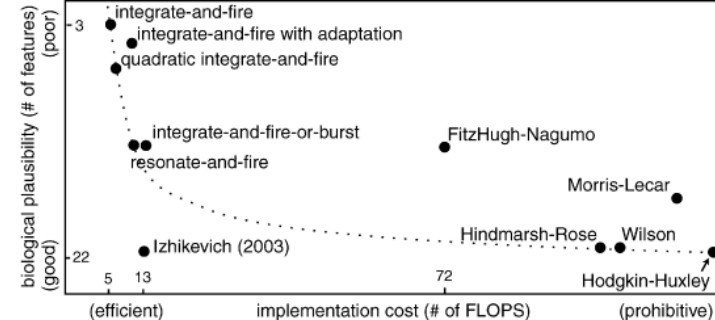
- An enhancement to the basic integrate-and-fire model is the integrate-and-fire-or-burst (I&FB) model proposed by Smith et al. [24].

$$v' = I + a - bv + gH(v - v_h)h(v_T - v)$$

if $v = v_{\text{thresh}}$, then $v \leftarrow c$

$$h' = \begin{cases} \frac{-h}{\tau^-}, & \text{if } v > v_h \\ \frac{(1-h)}{\tau^+}, & \text{if } v < v_h \end{cases}$$

- This model introduces a second variable, h , to describe the inactivation of the calcium T-current, enabling the representation of thalamo-cortical neurons.
- The dynamics of the T-current are governed by parameters g , v_h , v_T , τ^+ , and τ^- , with H being the Heaviside step function.
- The addition of this second variable allows for the emergence of bursting and other complex firing patterns, enhancing the model's biological realism.
- However, this increased biological fidelity comes at a computational cost, with simulations of the I&FB model requiring between 9 and 13 operations per 1 ms time step, depending on the membrane potential value.



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Morris-Lecar	+	+	+	-		-	-	+	+	+	+	+	+		+	+	-	+	+	-	600		
Wilson	-	+	+	+			+	+	+	+	+	+	+	+		+	+				180		
Hodgkin-Huxley	+	+	+	+			+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200		

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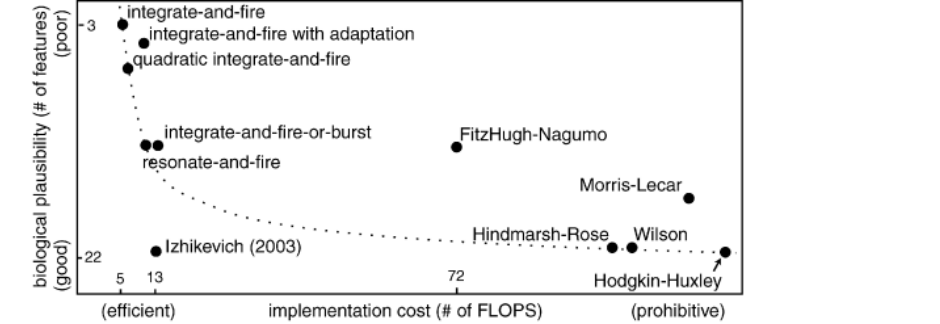
D. Resonate-and-Fire

- The resonate-and-fire neuron, introduced by Izhikevich [16], extends the basic integrate-and-fire model to a two-dimensional (2-D) framework.

$$z' = I + (b + \omega)z$$

if $\text{Im } z = a_{\text{thresh}}$, then $z \leftarrow z_0(z)$

- In this model, the membrane potential is represented by the real part of a complex variable z , with parameters b , ω , and a_{thresh} governing its dynamics.
- The resonate-and-fire model is characterised by its simplicity and efficiency, requiring only 10 operations to simulate 1 ms.
- When the frequency of oscillation ω is set to 0, the model behaves as an integrator, offering versatility in representing different neuronal dynamics.



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Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	600
Wilson	-	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

Spiking neuron models

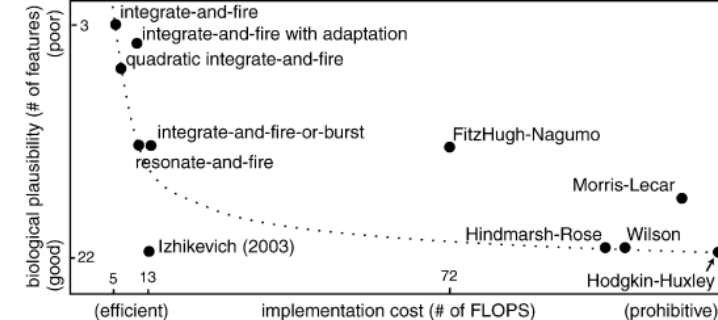
E. Quadratic I&F

- An alternative to the leaky integrate-and-fire (I&F) neuron is the quadratic I&F neuron, also known as the theta-neuron or the Ermentrout–Kopell canonical model.

$$v' = I + a(v - v_{\text{rest}})(v - v_{\text{thresh}})$$

$$\text{if } v = v_{\text{peak}}, \quad \text{then } v \leftarrow v_{\text{reset}}$$

- Described by the equation $v' = I + a(v - v_{\text{rest}})(v - v_{\text{thresh}})v_{\text{peak}}$, this model features a quadratic nonlinearity in its dynamics.
- With parameters v_{rest} and v_{thresh} representing resting and threshold values respectively, the model exhibits spike latencies and activity-dependent threshold behavior.
- Simulating the quadratic I&F neuron requires only seven operations per 1 ms, making it efficient for large-scale network simulations.
- Unlike its linear counterpart, this model offers bistability between resting and tonic spiking modes, providing a richer representation of neuronal dynamics.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	spike latency	subthreshold oscillations	resonator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
integrate-and-fire	-	+	-	-	-	-	+	-	-	-	-	+	-	-	-	-	-	-	-	-	5
integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	10
integrate-and-fire-or-burst	-	+	+	+	-	+	+	-	-	-	-	+	+	+	-	+	+	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	+	-	+	+	+	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	72
Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

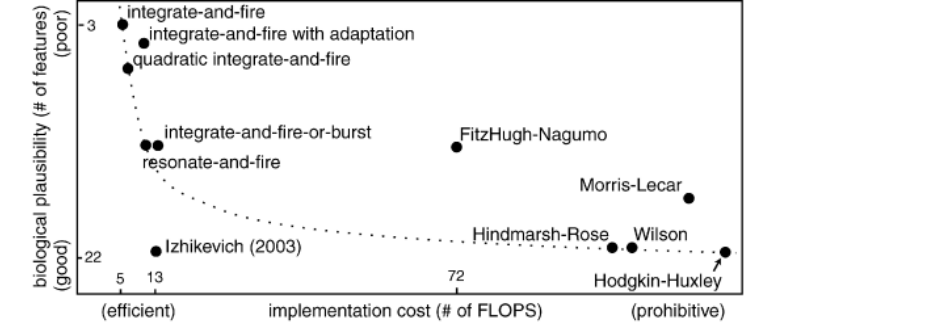
Spiking neuron models

F. Spiking Model by Izhikevich (2003)

- Proposed by Izhikevich, this spiking neuron model offers a versatile framework capable of replicating various firing patterns observed in cortical neurons.

$$\begin{aligned} v' &= 0.04v^2 + 5v + 140 - u + I \\ u' &= a(bv - u) \end{aligned} \quad \text{if } v \geq +30 \text{ mV,} \quad \text{then } \begin{cases} v \leftarrow c \\ u \leftarrow u + d. \end{cases}$$

- The model equations consist of membrane potential v and a recovery variable u , governed by a set of differential equations.
- After reaching a peak at +30 mV, the membrane potential and recovery variable are reset according to a defined rule, ensuring consistent spike magnitudes.
- With proper parameter selection, the model can emulate the firing behaviors of different types of cortical neurons.
- Simulating this model requires only 13 floating-point operations per 1 ms, making it efficient for large-scale cortical network simulations.
- Although the model can exhibit chaotic spiking with certain parameter configurations, maintaining numerical precision requires a small integration time step.
- Notably, the +30 mV value in the reset rule is not a fixed threshold but represents the peak of the spike, with the threshold value dynamically adjusted within a biologically plausible range.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	spike latency	subthreshold oscillations	resonator	integrator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
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integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	-	10
integrate-and-fire-or-burst	-	+	+	-	+	+	+	-	-	-	-	+	+	+	-	+	+	-	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	-	+	+	+	-	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	-	7
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FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	-	72
Hindmarsh-Rose	-	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

Spiking neuron models

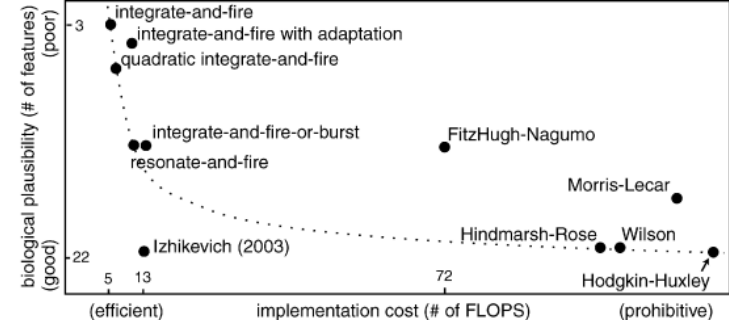
G. FitzHugh–Nagumo

- The FitzHugh–Nagumo model offers a simplified representation of spiking dynamics in resonator neurons by adjusting its parameters.

$$v' = a + bv + cv^2 + dv^3 - u$$

$$u' = \varepsilon(ev - u)$$

- Modelled by a set of ordinary differential equations, the parameters can be fine-tuned to replicate various neuro-computational properties, as depicted in Fig. 2.
- Due to the need to accurately simulate spike shapes, a relatively small time step, such as $\tau=0.25$ ms, is required.
- Simulation of this model demands 18 floating-point operations per 0.25 ms, translating to 72 operations per 1 ms.
- As it lacks a reset mechanism and consists of a 2-D system of ODEs, the FitzHugh–Nagumo model cannot autonomously exhibit chaotic dynamics or bursting.
- Introducing noise into this or similar 2-D models can enable stochastic bursting behavior.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	spike latency	subthreshold oscillations	resonator	integrator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
integrate-and-fire	-	+	-	-	-	-	+	-	-	-	-	+	-	-	-	-	-	-	-	-	-	5
integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	-	10
integrate-and-fire-or-burst	-	+	+	+	+	+	+	-	-	-	-	+	+	+	-	+	+	-	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	-	+	+	+	-	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	-	72
Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

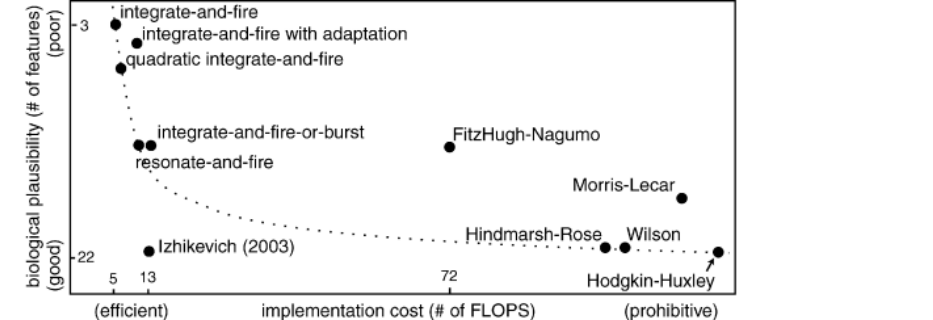
Spiking neuron models

H. Hindmarsh–Rose

- The Hindmarsh–Rose model, designed to simulate thalamic neurons, is expressed by a set of equations featuring functions F , G , and H .

$$\begin{aligned}v' &= u - F(v) + I - w \\u' &= G(v) - u \\w' &= \frac{(H(v) - w)}{\tau}\end{aligned}$$

- By selecting appropriate functions, the model theoretically has the capability to manifest all neuro-computational properties outlined in Fig. 1.
- However, determining these functions to accurately model specific neuron types, such as RS or LTS neurons, poses a significant challenge.
- Assuming polynomial functions of the third degree (ideally), the model requires a maximal time step of 0.25 ms to simulate action potential shapes.
- Each simulation step necessitates 30 floating-point operations per 0.25 ms, totaling 120 operations to simulate 1 ms of the model.
- This assessment remains optimistic and may face obstacles in practice, especially in achieving the desired level of accuracy.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	spike latency	subthreshold oscillations	resonator	integrator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
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integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	-	10
integrate-and-fire-or-burst	-	+	+	+	-	+	+	-	-	-	-	+	+	+	-	+	+	-	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	-	+	+	+	-	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	-	72
Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

Spiking neuron models

I. Morris–Lecar

- The Morris–Lecar model, proposed by Morris and Lecar, offers a simplistic 2-D framework initially intended to elucidate barnacle giant muscle fiber oscillations.

$$C\dot{V} = I - g_L(V - V_L) - g_{Ca}m_\infty(V) \times (V - V_{Ca}) - g_K n(V - V_K)$$

$$\dot{n} = \lambda(V)(n_\infty(V) - n)$$

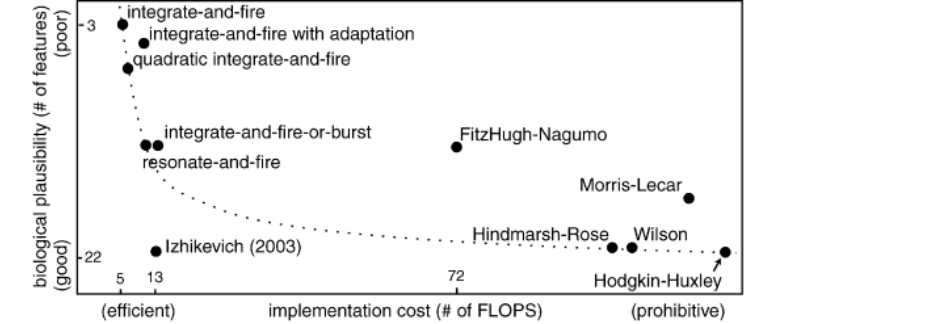
where

$$m_\infty(V) = \frac{1}{2} \left\{ 1 + \tanh \left[\frac{(V - V_1)}{V_2} \right] \right\}$$

$$n_\infty(V) = \frac{1}{2} \left\{ 1 + \tanh \left[\frac{(V - V_3)}{V_4} \right] \right\}$$

$$\lambda(V) = \bar{\lambda} \cosh \left[\frac{(V - V_3)}{(2V_4)} \right]$$

- Featuring a membrane potential equation with instantaneous Ca current activation and a slower K current activation equation, the model gained traction in the computational neuroscience community due to its biophysically meaningful parameters.
- The model's parameters allow for diverse spiking patterns, although achieving tonic bursting necessitates an additional equation, such as slow Ca current inactivation.
- Simulation of action potential shapes in the Morris–Lecar model mandates a time step significantly smaller than 1 ms for accurate results. A τ of 0.1 ms is found to yield reasonable outcomes in simulating cortical spiking neurons.
- With hyperbolic tangents and exponents involved, evaluating a single 0.1-ms time step requires approximately 60 floating-point operations, translating to 600 operations per 1 ms of simulation.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	spike latency	subthreshold oscillations	resonator	integrator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
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integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	-	10
integrate-and-fire-or-burst	-	+	+	+	-	+	+	-	-	-	-	+	+	+	-	+	+	-	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	+	+	+	-	-	+	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	-	72
Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

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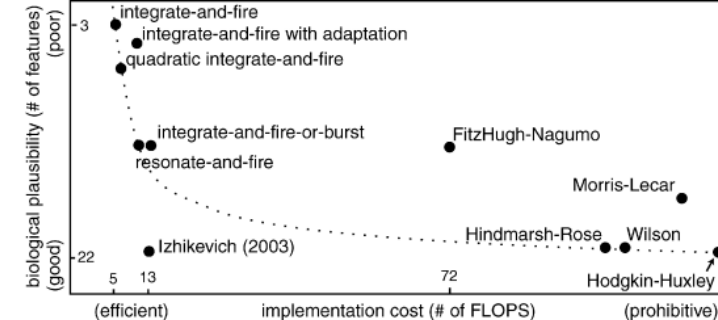
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integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	-	10
integrate-and-fire-or-burst	-	+	+	-	+	-	+	+	-	-	-	+	+	+	-	+	+	-	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	+	+	+	+	-	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	-	72
Hindmarsh-Rose	-	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

Spiking neuron models

J. Wilson Polynomial Neurons

- Wilson proposed a modeling approach using polynomial equations to capture the dynamics of cortical neurons.

$$I_j = g_j (V - E_j)$$

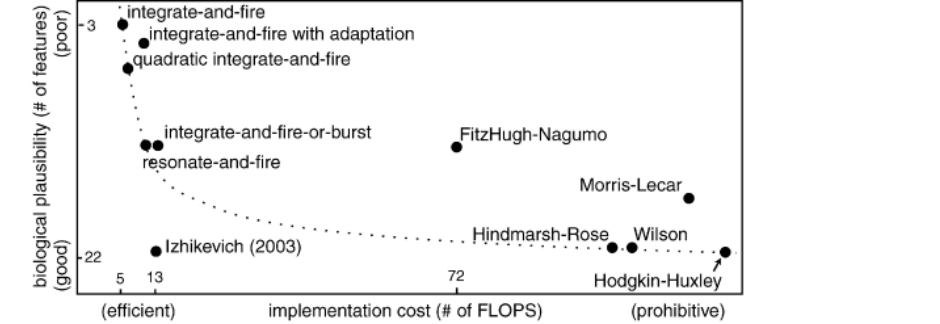
$$c \frac{dV}{dt} = -m_\infty(V) (V - E_{Na}) + g_K R (V - E_K) + I$$

$$\frac{dR}{dt} = \frac{1}{\tau_R} (-R + R_\infty(V))$$

$$m_\infty(V) = 17.8 + 47.6 + 33.8 V^2,$$

$$R_\infty(V) = 1.24 + 3.7V + 3.2 V^2.$$

- The model comprises four differential equations capable of exhibiting all neuro-computational properties outlined in Fig. 1, contingent upon suitable parameter selection—an inherently challenging task.
- A suggested time step for the model is 0.1 ms, although it can be extended up to 0.25 ms with minimal loss of precision or noticeable alteration of action potential shapes.
- Evaluation of the Wilson Polynomial Neurons model requires approximately 45 floating-point operations to assess 0.25 ms of simulation, equating to 180 operations per 1 ms.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	spike latency	subthreshold oscillations	resonator	integrator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
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integrate-and-fire with adapt.	-	+	-	-	-	-	+	+	-	-	-	+	-	-	-	-	+	-	-	-	-	10
integrate-and-fire-or-burst	-	+	+	-	+	-	+	+	-	-	-	+	+	+	-	+	+	-	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	-	+	+	+	-	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	-	+	-	-	+	+	-	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	-	+	+	-	+	+	-	-	72
Hindmarsh-Rose	-	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	+	-	+	+	-	-	600
Wilson	-	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

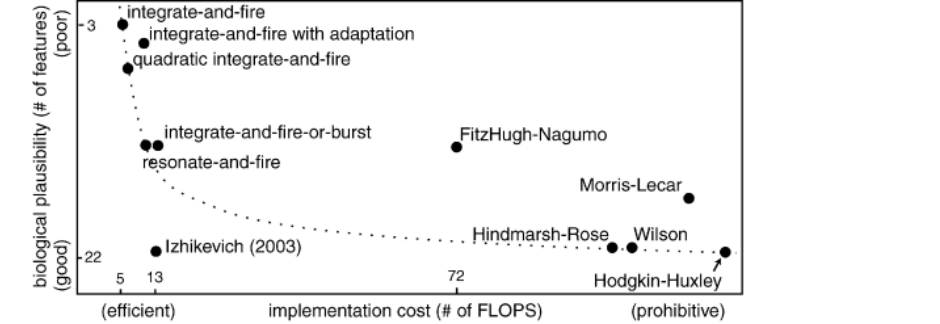
Spiking neuron models

K. Hodgkin–Huxley

- The Hodgkin–Huxley model, a cornerstone of computational neuroscience, comprises four equations and numerous parameters, omitted here, governing membrane potential and the activation and inactivation of sodium and potassium currents.

$$C_m \frac{dv}{dt} = -g_L(V - E_L) - g_{Na} m^3 h (V - E_{Na}) - g_K n^4 (V - E_K) + I$$

- While the model initially demonstrates limited behavior with default parameter values, it can manifest all properties depicted in Fig. 1 when parameters are appropriately adjusted.
- Conductance-based models, collectively referred to as Hodgkin–Huxley-type models, hold significance due to their biophysically meaningful and measurable parameters, enabling exploration of synaptic integration, dendritic cable filtering, dendritic morphology effects, and ionic current interactions.
- Implementation of the Hodgkin–Huxley model is computationally intensive, necessitating 120 floating-point operations for every 0.1 ms of model time, translating to 1200 operations per 1 ms. Consequently, its usage is viable for simulating only a limited number of neurons or in scenarios where simulation time is not a constraint.



Models	biophysically meaningful	tonic spiking	phasic spiking	tonic bursting	phasic bursting	mixed mode	spike frequency adaptation	class 1 excitable	class 2 excitable	subthreshold oscillations	resonator	rebound spike	rebound burst	threshold variability	bistability	DAP	accommodation	inhibition-induced spiking	chaos	# of FLOPS
integrate-and-fire	-	+	-	-	-	-	+	-	-	-	+	-	-	-	-	-	-	-	-	5
integrate-and-fire with adapt.	-	+	-	-	-	+	+	-	-	-	+	-	-	-	+	-	-	-	-	10
integrate-and-fire-or-burst	-	+	+	+	-	+	+	-	-	-	+	+	+	-	+	+	-	-	-	13
resonate-and-fire	-	+	+	-	-	-	+	+	-	+	+	+	+	-	+	+	+	-	+	10
quadratic integrate-and-fire	-	+	-	-	-	-	+	-	+	-	+	-	+	+	-	-	-	-	-	7
Izhikevich (2003)	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	13
FitzHugh-Nagumo	-	+	+	-	-	-	+	-	+	+	+	-	+	+	+	-	+	+	-	72
Hindmarsh-Rose	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	120
Morris-Lecar	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	-	+	+	-	600
Wilson	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	180
Hodgkin-Huxley	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	1200

Fig. 2. Comparison of the neuro-computational properties of spiking and bursting models; see Fig. 1. “# of FLOPS” is an approximate number of floating point operations (addition, multiplication, etc.) needed to simulate the model during a 1 ms time span. Each empty square indicates the property that the model should exhibit in principle (in theory) if the parameters are chosen appropriately, but the author failed to find the parameters within a reasonable period of time. (Izhikevich, 2004)

Cortical neuron conclusions

- Different models of spiking neurons serve different purposes. The Hodgkin–Huxley-type model is ideal for studying how neuronal behavior depends on physiological parameters. However, it's computationally expensive and limits real-time simulations to only a few neurons.
- On the other hand, the I&F model is efficient for simulating thousands of neurons in real time, but lacks key properties of cortical neurons. Similarly, the quadratic I&F model is efficient and captures some important properties but falls short in others.
- For understanding the temporal structure of cortical spike trains and studying spike-timing in information processing, Izhikevich's model is recommended. It's simple yet versatile, capable of exhibiting various firing patterns while remaining computationally efficient.

Noise in Spiking Neuron Models

- In vivo recordings of neuronal activity exhibit a high degree of irregularity, with spike trains of individual neurons appearing far from periodic.
- Deciphering whether this irregularity represents meaningful information or mere noise poses a significant challenge in neuroscience.

Experimental Evidence:

- Experiments suggest that neurons react reliably to fluctuating currents, with action potentials occurring consistently relative to stimulation.
- Nonstationary sensory stimulation also elicits similar spike trains upon repetition, indicating a structured response to certain stimuli.

Types of Noise:

- In spiking neuron models, noise is explicitly added to mimic the unpredictability of neuronal recordings.
- Three common implementations include escape noise, slow noise in parameters, and diffusive noise.

Subthreshold vs. Superthreshold Stimulation:

- Subthreshold stimulation relates to diffusive noise, while superthreshold stimulation can be linked to escape noise, affecting spike train variability differently.

Applications:

- The discussed noise models are applied to phenomena such as stochastic resonance, offering insights into the behavior of neuronal networks.

Comparison with Rate Models:

- Noise models are contrasted with rate models, highlighting their effectiveness in capturing the dynamics of neuronal activity.

Noise in Spiking Neuron Models

- In spiking neuron models like the Spike Response Model (SRM), noise is added to replicate the unpredictability of neuronal recordings.

Implementation of Noise:

- Three methods are presented: escape noise, slow noise in parameters, and diffusive noise.
- Subthreshold and superthreshold stimulation, and their implications for spike train variability, with a connection between diffusive and escape noise models.
- Application of noise models to stochastic resonance and comparison with rate models.

Experimental Evidence for Noise:

- Neuronal models driven by strong constant currents generate regular spike sequences, contrasting with the irregular behavior observed in vivo.
- Review of experiments demonstrating variability and irregularity in spike trains, raising questions about the origin of this behavior.

Are Neurons Noisy?

- In vivo experiments reveal noisy behavior of central neurons, with spike train variability observed even in the absence of external stimuli.
- Neuronal response to known intracellular input currents suggests deterministic behavior, but responses vary based on the nature of external stimuli.

Noise Sources

Intrinsic Noise Sources:

- **Thermal Noise:** Arises from the discrete nature of electric charge carriers, causing voltage fluctuations across electrical resistors (Johnson noise).
- **Ion Channel Fluctuations:** Stem from the finite number of ion channels in a patch of neuronal membrane, where fluctuations in channel conductance impact membrane potential.
 - Models considering finite ion channels replicate real neuron variability, especially with rapidly changing input currents.

Extrinsic Noise Sources:

- **Synaptic Transmission Failures:** Limit signal transmission within neuronal networks, with experiments suggesting only a fraction of presynaptic spikes lead to postsynaptic responses.
- **Network Effects:** Network connectivity and dynamics contribute to irregular neuronal spiking, even in the absence of intrinsic noise.
 - Excitatory and inhibitory networks with random connectivity produce highly irregular spike trains, influencing individual neuron activity.

Input-Dependent Renewal Systems

- Modeling Single Neuron Dynamics:
- Consider a neuron like an integrate-and-fire or SRM unit with known input current I and last firing time $t^{\wedge} < t$.
- The membrane potential u is determined by the input and last firing time.

$$u(t|\hat{t}) = \eta(t - \hat{t}) + \int_0^\infty \kappa(t - \hat{t}, s) I(t - s) ds, \quad (5.1)$$

$$u(t|\hat{t}) = u_r \exp\left(-\frac{t - \hat{t}}{\tau}\right) + \frac{1}{C} \int_0^{t - \hat{t}} \exp\left(-\frac{s}{\tau}\right) I(t - s) ds, \quad (5.2)$$

Predicting Next Action Potential:

- Without noise, the next firing time $t^{(f)}$ is determined by the threshold condition $u = \vartheta$.
- With noise, predicting exact firing time becomes probabilistic, aiming to calculate the probability distribution of the next firing time for time-dependent input I .

$$t^{(f)} = \min\{t > \hat{t} \mid u(t|\hat{t}) \geq \vartheta\}. \quad (5.3)$$

Input-Dependent Renewal Systems:

- Equations combined with stochastic spike generation represent input-dependent renewal systems.
- Renewal Processes: Describe sequential events (spikes) over time.
 - Central assumption: State (probability of next event) depends only on the time since the last event.
- broader Definition: State at time t depends on time since the last event $t^{\wedge}$ and input received since the last event.
 - Also termed as modulated renewal processes, nonstationary renewal systems, or inhomogeneous Markov interval processes.
 - Objective: Predict probability of next event given last event timing and received input.

Example: Light Bulb Failure as a Renewal System

- A typical example of a (potentially input-dependent) renewal system is a light bulb.
- Event: Failure of the bulb and its replacement.
- System state depends solely on the age of the current bulb, not on past bulbs.

Stationary Renewal Process:

- Usage pattern of bulbs is consistent (e.g., switched on for 10 hours nightly).

Non-Stationary Renewal Process:

- Usage varies (e.g., higher in winter, lower in summer, no usage during vacations).
- Aging of the bulb affected by frequency of switching.

Input-Dependent Renewal Theory:

- Utilised when tracking switching times.
- Input: Switching pattern.
- Objective: Calculate probability of next failure based on bulb age and switching pattern.

Interval Distribution

Estimating Interspike Interval (ISI) Distributions:

- Common method for studying neuronal variability under stationary input.
- Spike train of a single neuron recorded while driven by constant stimulus.
- Analysis involves plotting histogram of intervals between subsequent spikes.
- Histogram provides estimate of ISI distribution, denoted as $P_0(s)$ (Fig. 5.1A).

Generalising Interval Distributions:

- Focus on single neuron stimulated by known input current $I(t)$ and unknown noise.
- Question: Probability of next spike between t and $t + \Delta t$, given last spike at $\hat{t}$ and input $I(t')$ for $t' < t$?
- Probability density of firing $P_I(t|\hat{t})$ gives answer for $t \rightarrow 0$, with $\int_{t_1}^{t_2} P_I(t|\hat{t}) dt$ as probability of spike in $[t_1, t_2]$, given last spike at $\hat{t} < t_1$.
- Normalisation: $\int_0^\infty P_I(t|\hat{t}) dt = 1$, assuming P_I^{inact} (probability neuron stays inactive forever) vanishes.

Input-Dependent Interval Distribution:

- Denoted as $P_I(t|\hat{t})$.
- Conditioned on input $I(t')$ for $t' < t$.
- Called spike-triggered spike density, reflecting distribution of interspike intervals in presence of input current I (Fig. 5.1B).
- For renewal systems with stationary input, $P_I(t|\hat{t})$ reduces to $P_0(t - \hat{t})$.

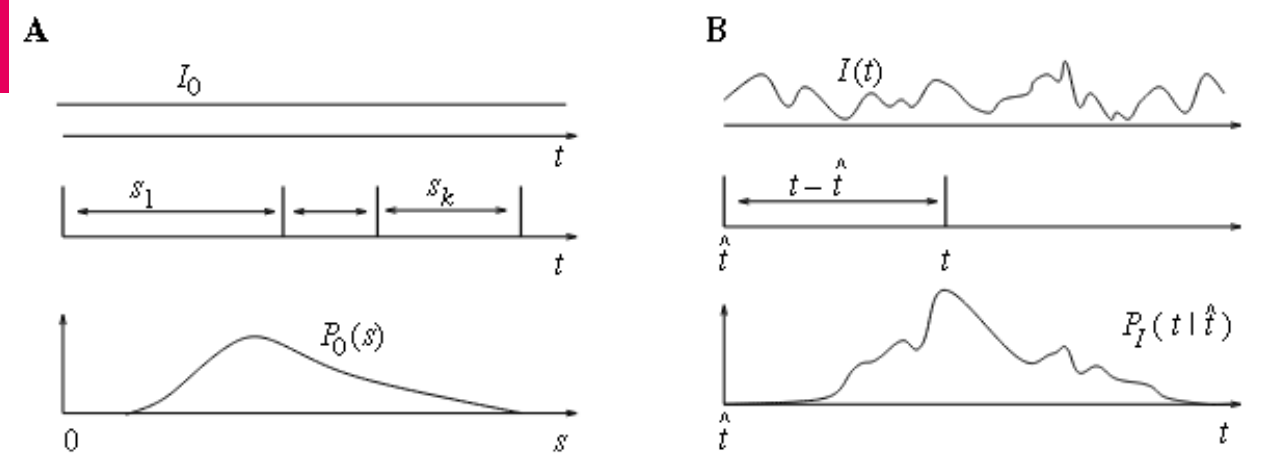


Fig. 5.1. **A.** Stationary interval distribution. A neuron is driven by a constant input (top). A histogram of the interspike intervals $s_1, s_2, \dots$ can be used to estimate the interval distribution $P_0(s)$ (bottom). **B.** Input-dependent interval distribution. A neuron, stimulated by the current $I(t)$ (top), has emitted a first spike at $\hat{t}$. The interval distribution $P_I(t|\hat{t})$ (bottom) gives the probability density that the next spike occurs after an interval $t - \hat{t}$.

$$\int_{\hat{t}}^{\infty} P_I(t|\hat{t}) dt = 1 - P_I^{\text{inact}} \quad (5.4)$$

Survivor Function and Hazard

Survivor Function:

- Interval distribution $P_I(t|\hat{t})$ integrates over time to yield probabilities.
- $\int_{\hat{t}}^t P_I(t'|\hat{t}) dt'$ is probability of neuron firing next action potential between $\hat{t}$ and t .
- $S_I(t|\hat{t}) = 1 - \int_{\hat{t}}^t P_I(t'|\hat{t}) dt'$ is survivor function: probability neuron "survives" from $\hat{t}$ to t without firing.
- Initial value: $S_I(\hat{t}|\hat{t}) = 1$, decreases to zero for $t \rightarrow \infty$.

Hazard:

- Rate of decay of $S_I(t|\hat{t})$ denoted by $\rho_I(t|\hat{t})$.
- In renewal theory, called "age-dependent death rate" or "hazard".

Relationship:

- Integration of Eq. (5.6) yields survivor function $S_I(t|\hat{t})$.
- Has intuitive interpretation: neuron must survive interval $(\hat{t}, t)$ without firing to emit next spike at t .
- Each of $\rho_I(t|\hat{t})$, $P_I(t|\hat{t})$, and $S_I(t|\hat{t})$ describes statistical properties of input-dependent renewal system.
- For stationary renewal systems, equations hold with replacement: equations (5.5)–(5.10) are standard results of renewal theory.

$$S_I(t|\hat{t}) = 1 - \int_{\hat{t}}^t P_I(t'|\hat{t}) dt' \quad (5.5)$$

$$\rho_I(t|\hat{t}) = -\frac{d}{dt} S_I(t|\hat{t}) / S_I(t|\hat{t}). \quad (5.6)$$

$$S_I(t|\hat{t}) = \exp \left[- \int_{\hat{t}}^t \rho_I(t'|\hat{t}) dt' \right]. \quad (5.7)$$

$$P_I(t|\hat{t}) = -\frac{d}{dt} S_I(t|\hat{t}) = \rho_I(t|\hat{t}) S_I(t|\hat{t}), \quad (5.8)$$

$$P_I(t|\hat{t}) = \rho_I(t|\hat{t}) \exp \left[- \int_{\hat{t}}^t \rho_I(t'|\hat{t}) dt' \right]. \quad (5.9)$$

$$\rho_I(t|\hat{t}) = -\frac{P_I(t|\hat{t})}{S_I(t|\hat{t})} = -\frac{P_I(t|\hat{t})}{1 - \int_{\hat{t}}^t P_I(t'|\hat{t}) dt'}. \quad (5.10)$$

$$P_I(t|\hat{t}) \rightarrow P_0(t - \hat{t}), \quad (5.11)$$

$$S_I(t|\hat{t}) \rightarrow S_0(t - \hat{t}), \quad (5.12)$$

$$\rho_I(t|\hat{t}) \rightarrow \rho_0(t - \hat{t}). \quad (5.13)$$

Example: from Interval Distribution to Hazard Function

- Suppose under stationary experimental conditions, interval distribution approximated as $P_0(s) \approx a_0$, where $a_0 > 0$ (Fig. 5.2A).

$$P_0(s) = \begin{cases} 0 & \text{for } s \leq \Delta^{\text{abs}} \\ a_0 (s - \Delta^{\text{abs}}) e^{-\frac{1}{2}a_0(s-\Delta^{\text{abs}})^2} & \text{for } s > \Delta^{\text{abs}} \end{cases} \quad (5.14)$$

$$\rho_0(s) = \begin{cases} 0 & \text{for } s \leq \Delta^{\text{abs}} \\ a_0 (s - \Delta^{\text{abs}}) & \text{for } s > \Delta^{\text{abs}} \end{cases} \quad (5.15)$$

Hazard Calculation:

- From Eq. (5.10), hazard is found to be:

$$\rho_1(s|t^\wedge) = \frac{P_0(s)}{S_1(s|t^\wedge)}$$

- Thus, during an interval after each spike, the hazard vanishes.
- Interpret Δ^{abs} as the absolute refractory time of the neuron.
- For $s > \Delta^{\text{abs}}$, hazard increases linearly:
Longer the neuron waits, higher its probability of firing.

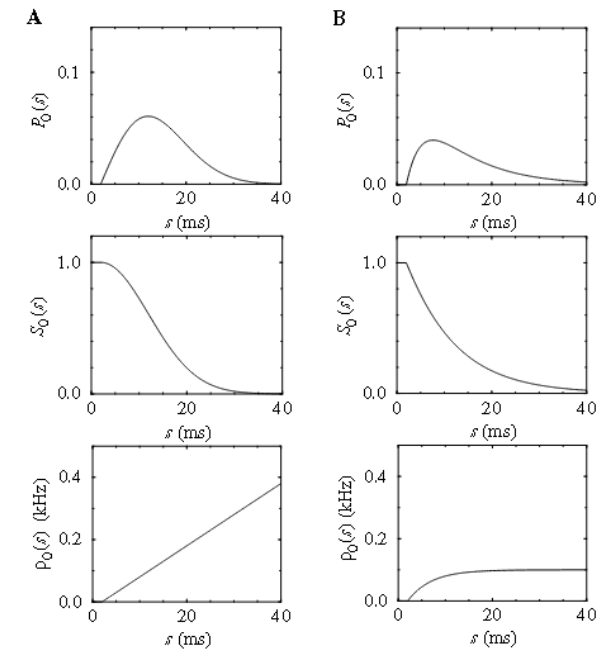


Fig. 5.2. **A.** Interval distribution $P_0(s)$ (top), survivor function $S_0(s)$ (middle) for a hazard function (bottom) defined by $\rho_0(s) = a_0 (s - \Delta^{\text{abs}}) \Theta(s - \Delta^{\text{abs}})$ with $a_0 = 0.01 \text{ ms}^{-2}$ and $\Delta^{\text{abs}} = 2 \text{ ms}$, where Θ is the Heaviside function. **B.** Similar plots as in A but for a hazard function defined by $\rho_0(s) = \nu \{1 - \exp[-\lambda(s - \Delta^{\text{abs}})]\} \Theta(s - \Delta^{\text{abs}})$ with $\nu = 0.1 \text{ kHz}$, $\lambda = 0.2 \text{ kHz}$, and $\Delta^{\text{abs}} = 2 \text{ ms}$.

Example: from Hazard Functions to Interval Distributions

- In auditory neurons of cats driven by stationary stimuli, the hazard function $\rho_0(t-t^*)$ increases after an absolute refractory time to a constant level (Goldberg et al., 1964).
- Time course of hazard function approximated as $\rho_0(s) = \lambda e^{-\nu(s-\Delta^{\text{abs}})}$ with parameters Δ^{abs} , λ , and ν (Fig. 5.2B).

$$\rho_0(s) = \begin{cases} 0 & \text{for } s \leq \Delta^{\text{abs}} \\ \nu [1 - e^{-\lambda(s-\Delta^{\text{abs}})}] & \text{for } s > \Delta^{\text{abs}} \end{cases} \quad (5.16)$$

$$S_0(s) = \begin{cases} 1 & \text{for } s \leq \Delta^{\text{abs}} \\ e^{-\nu(s-\Delta^{\text{abs}})} e^{\nu \rho_0(s)/\lambda} & \text{for } s > \Delta^{\text{abs}} \end{cases} \quad (5.17)$$

Interpretation:

- Relationship between the hazard function (5.16) and neuronal dynamics.

Calculation:

- Given the hazard function, we can calculate the survivor function and interval distributions.
- Applying Eq. (5.7) yields: $S_0(s) = e^{-\Lambda(s)}$, where $\Lambda(s) = \int_0^s \rho_0(s') ds'$.
- Interval distribution is given by: $P_0(s) = \rho_0(s) S_0(s)$.

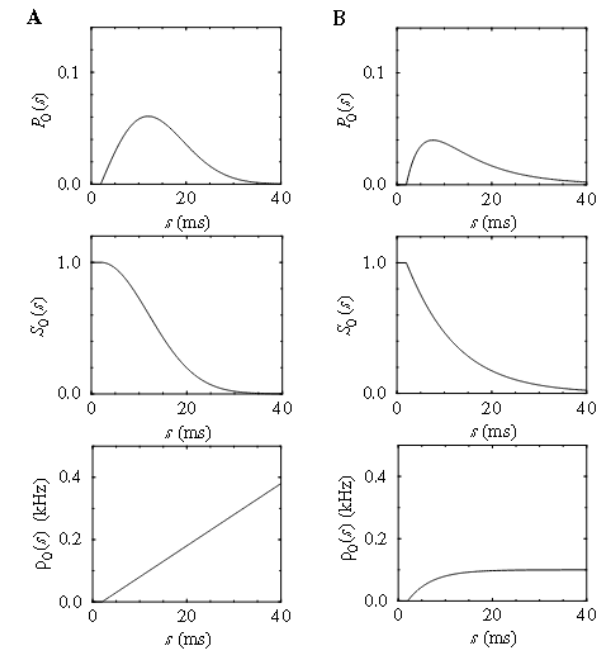


Fig. 5.2. **A.** Interval distribution $P_0(s)$ (top), survivor function $S_0(s)$ (middle) for a hazard function (bottom) defined by $\rho_0(s) = a_0 (s - \Delta^{\text{abs}}) \Theta(s - \Delta^{\text{abs}})$ with $a_0 = 0.01 \text{ ms}^{-2}$ and $\Delta^{\text{abs}} = 2 \text{ ms}$, where Θ is the Heaviside function. **B.** Similar plots as in A but for a hazard function defined by $\rho_0(s) = \nu [1 - \exp[-\lambda(s - \Delta^{\text{abs}})]] \Theta(s - \Delta^{\text{abs}})$ with $\nu = 0.1 \text{ kHz}$, $\lambda = 0.2 \text{ kHz}$, and $\Delta^{\text{abs}} = 2 \text{ ms}$.

Example: Poisson Process

Comparison to Poisson Process:

- Let's compare the hazard functions of previous examples to a homogeneous Poisson process generating spikes stochastically at a fixed rate ν .
- For a Poisson process, the hazard $\rho_0(s)$ is constant ($\rho_0(s) \equiv \nu$), with no dependence on previous spikes.
- From Eq. (5.8), the survivor function is $S_0(s) = \exp[-\nu s]$.
- Interval distribution is exponential: $P_0(s) = \nu \exp[-\nu s]$.
- Interval distribution and survivor function of a Poisson neuron with constant rate ν are plotted in Fig. 5.3A.
- Note: The interval distribution peaks at $s=0$, indicating high likelihood of extremely short intervals.

Incorporating Refractoriness:

- Real neurons exhibit refractoriness, unlike a Poisson process where $P_0(s)$ peaks at $s=0$.
- To incorporate refractoriness, a modified Poisson process is defined with a hazard function:

$$\rho_0(s) = \begin{cases} 0 & \text{for } s < \Delta^{\text{abs}} \\ \nu & \text{for } s > \Delta^{\text{abs}} \end{cases} \quad (5.19)$$

- This process generates spike trains with an interval distribution (Fig. 5.3B).

Comparison:

- Compare hazard function of the Poisson neuron with absolute refractoriness (Eq. 5.19) to more realistic hazard (Eq. 5.16).
- Main difference: Hazard in Eq. (5.19) jumps from absolute refractoriness to constant firing rate, while in Eq. (5.16) the transition is smooth.

$$P_0(s) = \nu e^{-\nu s} \quad \text{for } s > 0. \quad (5.18)$$

$$P_0(s) = \begin{cases} 0 & \text{for } s < \Delta^{\text{abs}} \\ \nu \exp[-\nu(s - \Delta^{\text{abs}})] & \text{for } s > \Delta^{\text{abs}} \end{cases} \quad (5.20)$$

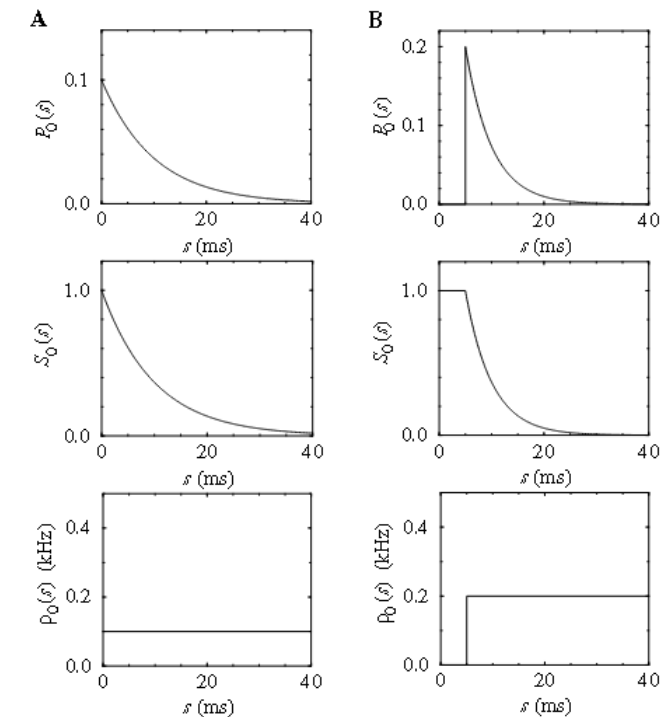


Fig. 5.3. Interval distribution $P_0(s)$ (top), survivor function $S_0(s)$ (middle), and hazard function (bottom) for a Poisson neuron (A) and a Poisson neuron with absolute refractoriness ($\Delta^{\text{abs}} = 5$ ms) (B).

Stationary Renewal Theory and Experiments

Renewal Theory and Stationary Input:

- Renewal theory is typically associated with stationary input conditions.
 - Interval distribution P_0 can be estimated experimentally from a single long spike train.
 - Applicability of renewal theory relies on the hypothesis that memory back to the last spike suffices to describe spike statistics.
 - No correlation between one interval and the next is assumed.
- Intracellular recordings under in vitro conditions allow direct testing with constant input current.
 - In vivo conditions pose challenges due to uncertain and non-stationary input to neurons within neural systems.

Testing the Renewal Hypothesis:

- Renewal hypothesis can be tested experimentally by measuring correlation between subsequent intervals.
 - Small correlations suggest spiking can be described as a stationary renewal process.
- Assuming renewal hypothesis holds, experimental P_0 allows calculation of survivor function S_0 and hazard ρ_0 via equations (5.5) and (5.10).
 - Additional assumptions about noise enable interpreting the form of hazard $\rho_0(t|t^*)$ in terms of neuronal dynamics.
 - Reduced hazard after a spike indicates neuronal refractoriness.

Challenges in Experimental Translation:

- Stationary input conditions are a mathematical concept, not easily translated into experiments.

Utilising Experimental Data:

Statistical Information:

- For a stationary renewal process, P_0 contains statistical information such as mean firing rate, autocorrelation function, and noise spectrum.

Mean Firing Rate

- The power spectrum (or power spectral density) of a spike train is defined as $P(\omega) = \lim_{T \rightarrow \infty} P_T(\omega)$, where P_T is the power of a segment of length T of the spike train
- The power spectrum $P(\omega)$ of a spike train is equal to the Fourier transform $\hat{C}_{ii}(\omega)$ of its autocorrelation function
- In the limit of $T \rightarrow \infty$, Eq. (5.25) becomes identical to Eq. (5.26) so that the assertion follows.
- The power spectral density of a spike train during spontaneous activity is called the noise spectrum of the neuron

$$P_T(\omega) = \frac{1}{T} \left| \int_{-T/2}^{T/2} S_i(t) e^{-i\omega t} dt \right|^2. \quad (5.25)$$

$$\begin{aligned} \hat{C}_{ii}(\omega) &= \int_{-\infty}^{\infty} \langle S_i(t) S_i(t+s) \rangle e^{-i\omega s} ds \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} S_i(t) \int_{-\infty}^{\infty} S_i(t+s) e^{-i\omega s} ds dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} S_i(t) e^{+i\omega t} dt \int_{-\infty}^{\infty} S_i(s') e^{-i\omega s'} ds'. \end{aligned} \quad (5.26)$$

Escape Noise

- Various methods to introduce noise in spiking neuron models.
- Focus on the "noisy threshold" or escape model.

Concept:

- Neuron can fire without reaching the formal threshold (ϑ).
- May remain quiescent even after surpassing the threshold.

Escape Rate:

- Introduced to model the probability of firing or staying quiescent.
- Depends on the current state of the neuron.

Escape Rate and Hazard Function

- Traditional models often use a deterministic threshold for firing.
- In the escape model, stochastic firing criterion replaces the strict threshold.
- Spikes can occur at any time with a probability density f , the escape rate.

Concept:

- f acts as an escape rate, similar to chemical reaction models.
- In neuron models, f is referred to as firing intensity or stochastic intensity.
- Time Dependency:
 - Membrane potential $u(t|\hat{t})$ determines f , making it time-dependent.
 - f translates $u(t|\hat{t})$ into the hazard function $\rho_I(t|\hat{t})$.
- Generalisation:
 - Escape rate can depend not only on membrane potential u but also on its derivative u' .
 - Choice of escape function f is arbitrary but typically approaches zero as u tends to negative infinity.

Examples:

- Various forms of f can be considered, accounting for different spike initiation mechanisms.
- The discussion of more complex escape rate models will be addressed later.

$$\rho = f(u - \vartheta), \quad (5.41)$$

$$\rho_I(t|\hat{t}) = f[u(t|\hat{t}) - \vartheta], \quad (5.42)$$

$$\rho_I(t|\hat{t}) = f[u(t|\hat{t}), \dot{u}(t|\hat{t})]. \quad (5.43)$$

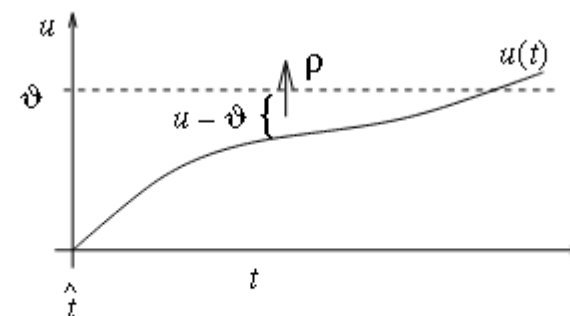


Fig. 5.5. Noisy threshold. A neuron can fire at time t with probability density $\rho[u(t) - \vartheta]$ even though the membrane potential u has not yet reached the threshold ϑ .

Example: Hard and Soft Threshold

- Escape model introduced as a noisy alternative to deterministic threshold models.
- Explore the relationship between noisy and deterministic threshold models.

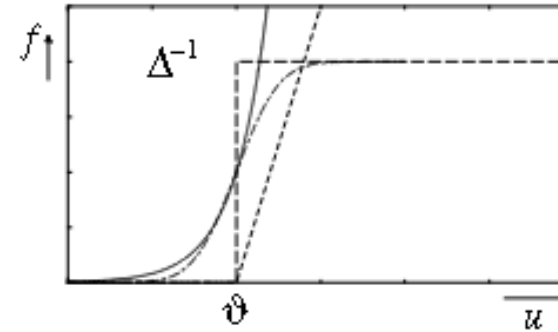


Fig. 5.6. Soft threshold escape rates. Exponential function (solid), piecewise linear function (dotted), step function (dashed), and error function (dot-dashed). The step function and error function saturate at a maximum rate of Δ^{-1} . The threshold is ϑ .

Softening the Threshold:

- Soft threshold models allow for a gradual increase in firing intensity as membrane potential approaches threshold.
- Different models capture various aspects of neuronal behavior and have different applications in describing spike generation.

Example: Motivating a Sigmoidal Escape Rate

Consider a model with discrete time steps $t_n = n$.

- After each time step, a new threshold value θ' is chosen from a Gaussian distribution with mean θ .
- Firing probability at time step t_n equals the probability that θ' is below the membrane potential $u(t_n)$.
- Firing probability divided by step size interpreted as firing intensity.

Alternative Approach:

- Study a model with fixed threshold and stochastic membrane potential.
- Total membrane potential at time step t_n : $u(t_n) + u(t_n - 1)$, where u is deterministic and u is stochastic.
- If u is chosen independently from a Gaussian distribution with variance σ and mean 0 at each time step, Eq. (5.47) is obtained again.

Issues and Considerations:

- Low-noise Limit: Recovery of deterministic threshold model is problematic for $\sigma \rightarrow 0$.
- Discretization Dependency: Simulations may depend on discretization Δt because the firing intensity diverges as $\Delta t \rightarrow 0$.
- Despite its problems, sigmoidal escape rate is used in continuous-time neuronal models, motivated by Gaussian distribution of threshold values or membrane potentials with band-limited noise.
- Time scale in continuous-time models should be proportional to the correlation time τ_{corr} of the noise in the membrane potential.

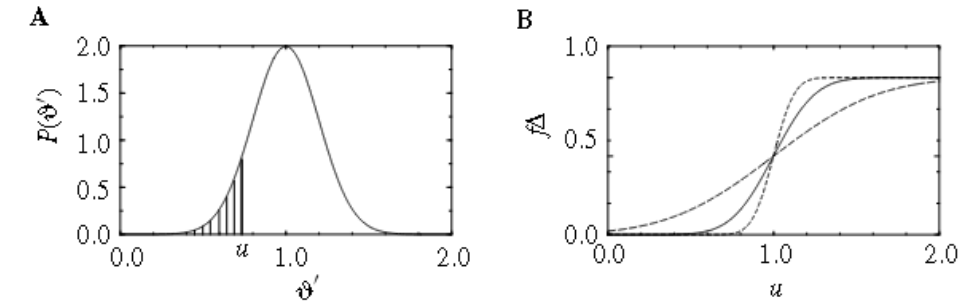


Fig. 5.7. **A.** Gaussian distribution of thresholds $P(\theta')$ with mean $\theta = 1$ and variance $\sigma = 0.2$. The shaded area gives the probability that u is above threshold. **B.** Escape rate according to Eq. (5.51) with $\theta = 1$ for different noise levels $\sigma = 0.1$ (dotted line), $\sigma = 0.2$ (solid line), and $\sigma = 0.5$ (dashed line).

Example: Transition from Continuous to Discrete Time

- Begin with a model in continuous time and transition to discrete time, commonly done in simulations.
- Previous example showed limitations of directly going from discrete to continuous time.

Discretization Scheme:

- In a straightforward discretisation, firing probability during a time step t is calculated as $\rho_i(t|\mathbf{t}^*)t$.

Improved Discretization:

- Calculate the probability that a neuron does not fire in a time step t .

Advantages:

- Even if β diverges as $u \rightarrow \infty$, the probability remains bounded between zero and one.
- Increasing the discretization Δt mainly shifts the firing curve to the left, while maintaining the general form.
- Higher noise levels result in flatter curves.

- Fig. 5.8A: Increasing Δt shifts the firing curve.
- Fig. 5.8B: Increased noise level flattens the curve.

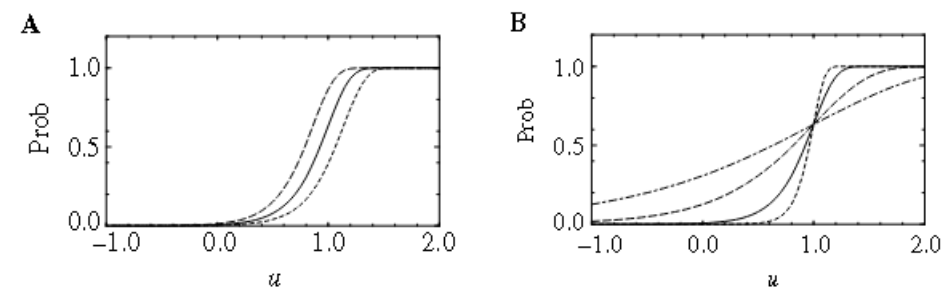


Fig. 5.8. **A.** Probability of firing in a discrete time interval Δt as a function of the membrane potential u for different discretizations $\Delta t = 0.5$ ms (dashed line), $\Delta t = 1$ ms (solid line), and $\Delta t = 2$ ms (dotted line) with $\beta = 5$. **B.** Similar plot as in **A** but for different noise levels $\beta = 10$ (dotted line), $\beta = 5$ (solid line), $\beta = 2$ (dashed line), and $\beta = 1$ (dot-dashed line) with $\Delta t = 1$ ms. The escape rate is given by Eq. (5.45) with parameters $\vartheta = 1$ and $\tau_0 = 1$ ms.

Interval Distribution and Mean Firing Rate

- Combining Escape Rate Model with Renewal Theory:
- This section integrates the escape rate model with renewal theory to calculate the input-dependent interval distribution $P(t|t^*)$ for escape rate models.

Comparison to Stationary Poisson Process:

- Compare the expression with the interval distribution of the stationary Poisson process.
- Key difference: Hazard $p_i(t|t^*)$ depends on both the last firing time t^* and the potentially time-dependent input.

Role of Refractoriness:

- Refractoriness strongly influences the interval distribution of neurons.
- Escape models demonstrate the relationship between the hazard $p_i(t|t^*)$ and refractoriness.

- Expressing p_i by the Escape Rate:
- Assume the escape rate f is a function of membrane potential u only.

- Figure 5.9 illustrates the interval distribution (Eq. 5.56) for constant input current I_0 as a function of $s = t - t^*$.
- With the normalization $\int_0^\infty \kappa(s) ds = 1$, $h_0 = I_0$.
- Refractory term η makes short intervals impossible, with the maximum occurring at a finite value of s .
- Increasing I_0 shifts the maximum to the left.

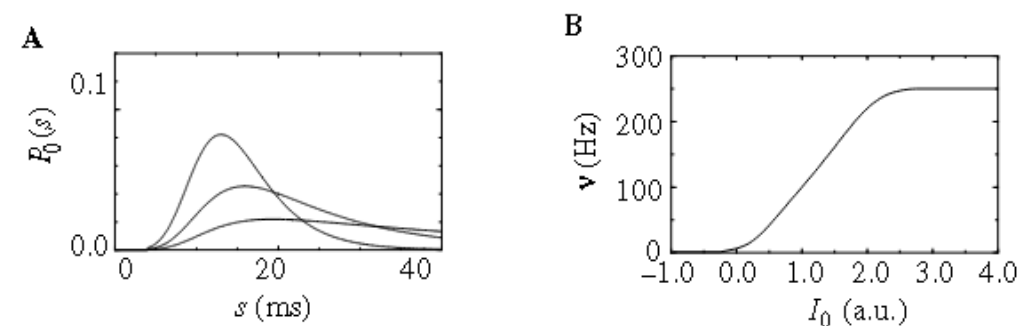


Fig. 5.9. **A.** Interval distribution $P_0(s)$ for a SRM_0 neuron with absolute refractory period $\Delta^{\text{abs}} = 4$ ms followed by an exponentially decreasing afterpotential as in Eq. (5.63) with $\eta_0 = 1$ and $\tau = 4$ ms. The model neuron is stimulated by a constant current $I_0 = 0.7, 0.5, 0.3$ (from top to bottom, arbitrary units). **B.** Output rate ν as a function of I_0 (gain function). The escape rate is given by Eq. (5.45) with $\vartheta = 1$, $\beta = 5$, and $\tau_0 = 1$ ms.

Example: periodic input

- Study the model SRM_0 with periodic input $I(t)=I_0+I_1\cos(\omega t)$, where ω is the modulation frequency.
- This leads to an input potential $h(t)=h_0+h_1\cos(\omega t+\phi_1)$ with bias $h_0=I_0$ and a periodic component with amplitude h_1 and phase ϕ_1 .

Choice of Refractory Kernel and Escape Rate:

- Adopt a refractory kernel with absolute and relative refractoriness defined as aa and adopt the exponential escape rate.
- Calculation of Spike Probability:
- Suppose a spike has occurred at $t^*=0$. The probability density that the next spike occurs at time t is given by $P_I(t|t^*)$.
- The result is shown in Fig. 5.10. The periodic component of the input is well represented in the response of the neuron.

Impact of Noise:

- Emphasise that the threshold is at $\theta=1$. Without noise, there would be no output spike.
- At very high noise levels, the modulation of the interval distribution would be much weaker.
- A certain amount of noise is beneficial for signal transmission.

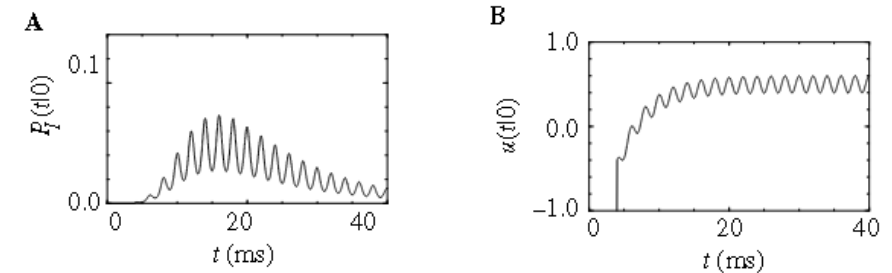
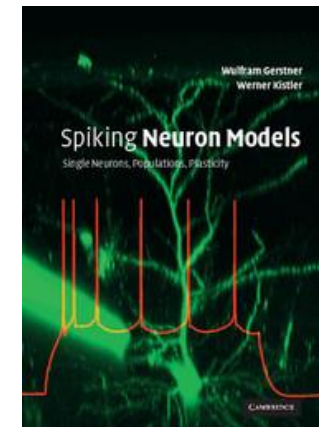


Fig. 5.10. **A.** Input-dependent interval distribution $P_I(t|0)$ for a SRM_0 neuron as in Fig. 5.9 stimulated by a periodically modulated input field $h(t) = h_0 + h_1 \cos(2\pi f t)$ with $h_0 = 0.5$, $h_1 = 0.1$ and frequency $f = 500$ Hz. **B.** The membrane potential $u(t|0) = \eta(t) + h(t)$ during stimulation as in **A.**

References

- [1] Gerstner, W. and Kistler, W.M., 2002. Spiking neuron models: Single neurons, populations, plasticity. Cambridge university press.
- [2] E. M. Izhikevich, "Which model to use for cortical spiking neurons?," in *IEEE Transactions on Neural Networks*, vol. 15, no. 5, pp. 1063-1070, Sept. 2004, doi: 10.1109/TNN.2004.832719.





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University

Department of Computer Science

Summary

In this lecture you learnt to:

- Cortical Neurons Properties
- Suitable Neurons for Cortical Responses
- Spike Train Variability
- Are Neurons Noisy?
- Noise Sources
- Statistics of Spike Trains

This lecture contributes to the following learning outcomes:

MLO2 - Demonstrate awareness of the different main paradigms corresponding to each of the several disciplines involved (computer science, digital electronics, AI), to be able to contextualise cognitive systems research and its current outlook.

MLO4. Understand and explain the applications areas of ACS, and the consequent social and ethical issues.

MLO6. Demonstrate in-depth skills in system architecture design and programming required to implement ACS.

MLO08. Demonstrate the capability of testing applications targeting real applications.

MLO10. Demonstrate the use of Distributed Database systems to handle data scalability and considering costs, benefits, risks and relevant business goals.

- Escape Noise



Nottingham Trent
University

Department of Computer Science

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Lecture 7: Cortical and Noisy Neurons

Pedro Machado pedro.machado@ntu.ac.uk